

RITESH YADAV

Generative AI & Full-Stack Software Engineer

Noida, UP, India | [+91-8434041818](tel:+91-8434041818) | ratohikumar@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

TECHNICAL SKILLS

Generative AI & LLMs	OpenAI (ChatGPT API), Google Gemini, RAG Architecture, Vector DBs, Semantic Search, Virtual Try-On (VTON) AI, AI Background Removal, MCP (Model Context Protocol), Voice AI
Languages & Frameworks	JavaScript (ES6+), TypeScript, Python, Node.js, Express.js, React.js, Next.js 16, FastAPI, PHP, C++, SQL
Databases & Distributed	RDBMS (MySQL, PostgreSQL), MongoDB, Redis (In-Memory Caching), RabbitMQ, Microservices, REST & GraphQL
DevOps & Tooling	Docker, Kubernetes, AWS (EC2), Linux, Git, GitLab CI/CD, GitHub Actions, Playwright (via MCP), Stripe, Razorpay

EXPERIENCE

Webkul

Software Engineer (Software Analyst)

Feb 2025 – Present

Noida, UP, India (Onsite)

- Engineered enterprise multilingual conversational AI chatbots powered by **OpenAI (ChatGPT API)** and **Google Gemini API**, designing an in-house **RAG pipeline with Vector Databases** for semantic search over store catalogs for context-aware, grounded responses.
- Integrated multimodal Generative AI capabilities for e-commerce, including automated **AI image background removal** pipelines and **Virtual Try-On (VTON) AI integrations** to enable interactive, real-time product previews.
- Automated end-to-end regression testing workflows by leveraging **MCP (Model Context Protocol)** and **Playwright** within GitLab CI/CD, enabling agentic test execution and self-healing test pipelines.
- Architected custom connectors for **Amazon SP-API**, **eBay REST/GraphQL APIs**, **Zoho**, and **Wix** on CS-Cart and OpenCart, building **RabbitMQ** background queues with single-consumer locks to sync 50,000+ daily events.
- Optimized high-volume **RDBMS** database queries (MySQL) and designed composite indexes on mapping tables, leveraging **Redis caching** to eliminate sync bottlenecks and race conditions across 100K+ SKUs; integrated **Stripe Custom Accounts** and **Veri*Factu** tax compliance.

Refresh Infratech Pvt. Ltd.

Software Developer (Part-Time)

Oct 2022 – Apr 2023

Onsite

- Built a contactless QR-based restaurant ordering system enabling real-time digital menu browsing, table-side ordering, and instant checkout using **Stripe Payments**.
- Developed real-time backend workflows and order routing with **EJS** templating and Node.js/Express, deploying the platform live for automated dine-in experiences.

PROJECTS

Shramik v2 [[Live](#) | [GitHub](#)] | *Next.js 16, React 19, TypeScript, MongoDB, Voice AI*

Dec 2024 – Present

- Re-architected platform to **Next.js 16 (App Router)** and **TypeScript**, integrating **Voice AI** job matching to empower low-literacy informal workers with conversational search and spoken command workflows.
- Engineered **e-Shram / OTP verification**, a double-entry **ledger transaction engine** with **Razorpay escrow** payouts, and secured backend routes using **Zod** validation and stateless **JWT** auth.

EduYug — AI-Augmented Course Platform [[GitHub](#)] | *NestJS 10, Next.js 14, pgvector, Redis, BullMQ*

May 2025

- Architected a modular monolith in Turborepo with **NestJS 10**, **Next.js 14**, and **PostgreSQL 16**, embedding native semantic search via **pgvector** cosine similarity.
- Engineered a timestamped **RAG AI Tutor** with millisecond video seeking, an adaptive bitrate **HLS.js player** with BullMQ/FFmpeg transcoding, and an immutable **double-entry ledger** with Razorpay.

EDUCATION

G.L. Bajaj Institute of Technology & Management

MCA - Master of Computer Applications — Grade: 7.4 CGPA

Greater Noida, UP, IN

Oct 2023 – Jun 2025

St. Xavier College Ranchi

B.Sc. in Information Technology — Grade: 7.0 CGPA

Ranchi, Jharkhand, IN

Oct 2020 – May 2023